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Brand-usage networks: a demand side approach to brand architecture

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Brands represent the outcomes of marketing investment and are an important component of the market value of the firm. Managing a brand portfolio requires an understanding of the role or function of each brand; this critical analysis yields the brand architecture. Typically this involves supply side analysis of a company's owned brands. Using an approach modeled on market structure analysis, we propose simultaneously capturing the interdependence of brands within the portfolio as well as competitive positioning. In this paper we advocate the use of network analysis to provide metrics on brand relationships in the market place. This demand side approach examines the interaction between brands and usage, and provides behavioral insights into the brand architecture that can be used to develop brand portfolio strategies. An example using data from the US car rental industry demonstrates brand-usage interactions as separate and distinct entities in a relational structure. Strategic utility is demonstrated by using network analysis to define the Dollar Thrifty Automotive Group brand architecture.

Keywords: brand-usage; brand architecture; brand portfolio; network analysis; market structure analysis; consumer behavior

Introduction

Many companies own multiple brands, some of which have been internally created or externally acquired through mergers and acquisitions. Though some brands provide diversification across unrelated businesses, other brands compete in the same or overlapping product categories and may provide the company with multiple alternatives in a single consumer segment. Since brands may represent a significant portion of the market value or assets of the company (Madden, Fehle, & Fournier, 2006; also see Salinas & Ambler, 2008), defining an architecture and managing the brand portfolio is of critical concern for creating and executing a company's business strategy.

Brand architecture is an 'organizing structure of the brand portfolio that specifies brand roles and the nature of relationships between brands' (Aaker & Joachimsthaler, 2000b, p. 8). Much of the existing literature on managing the brand portfolio has focused on brand architecture as a classification system for different types of portfolios (Aaker, 2004; Aaker & Joachimsthaler, 2000a; Strebringer, 2003). The spectrum of a 'branded house' to a 'house of brands' prescribes architectures and identifies the organization and management of multiple brands and the brand hierarchy (Keller, 2008). While this is an

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important contribution, it has been largely a *supply side* approach that stresses the relationship of a company to its brands. But because brands also represent a relationship between the company and consumers, it should be important to understand the brand architecture from a customer's perspective, an approach that requires *demand side* analysis (Fournier, 1998).

In this article, we briefly examine the prevailing supply side approach to identifying brand architecture, discuss the value of demand side analysis for improving the utility of brand architecture, and propose the application of network analysis as an effective demand side method that will enhance the use of brand architecture for strategy formation.

Brand architecture and portfolio management

Supply side perspective: prescriptive structure

Brand architecture is a strategic tool for achieving brand coherence (Balmer, 1998) and creating rational marketing strategy. Most large companies (e.g. those included in the Asiaweek 1000, the FT 500-Europe, and the Fortune 500) have portfolios with multiple brands. For example, Morgan and Leotte do Rego's (2006) sample of 149 Fortune 500 Companies found that each had over 18 brands, with the maximum being 400 brands. Logically, prioritizing the relative 'importance' and identifying the strategic roles of these brands is required for resource allocation.

Supply side analyses support management strategy and focus on how a firm competes. Supply side methods may include qualitative case studies, brand audits, and key-informant interviews as well as quantitative examination of company databases. For example, Kumar (2003, p. 89) suggests the way to refine the role of the brand in the portfolio is a brand audit listing a brand's market share, geographic presence, percentage of sales and profits, and noting whether each brand generates or uses up cash. These measures allow management to assign a strategic role for each brand – cost leadership, differentiation, or focus – and allocate resources depending on industry attractiveness and each brand's competitive strength (Porter, 1980). The set of role relationships defines a company's brand architecture.

To understand brand roles, Aaker (2004, p. 27) recommends relying on tools that can show the brand groupings, hierarchies, and networks. Empirical supply side research on brand portfolios reinforces a critical link between the portfolio and firm performance by modeling and testing linkages using data gathered from a firm (Morgan & Leotte do Rego, 2006; Rao, Agarwal, & Dahlhoff, 2004; Simmons, Bickart, & Buchanan, 2000). An important limitation in the literature is that this research essentially maps each brand owned onto one of several prescribed portfolio structures such as Kapferer's (1992) hierarchy of brands or Aaker's (1996) brand systems framework, which incorporates the 'house of brands' to 'branded house' spectrum.

As described by Rajagopal and Sanchez (2004), the supply side view of brand architecture is, 'a legacy of past management decisions' and competitive realities that result in brand baggage. Fundamentally, the supply side approach seems to eschew the notion of marketing as an exchange process creating value for *both* the firm *and the consumer* (Bagozzi, 1975). From this supply side perspective, management is encouraged to treat competing brands as external factors, or ignore them altogether, when considering a company's own brand portfolio. Thus one key question usually left unasked is: how do competitors' brands interact with and influence owned brands? This shortcoming of supply side analysis underscores the need for counter-balancing research on brand architecture that looks at the structure (architecture) from a consumer demand perspective.

Demand side perspective: insights from marketing structure analysis

Demand side analysis focuses on where a brand competes – consumer segments, locations, and product types – as well as on what the consumer wants or needs and how the consumer will use the product (Grant, 2010). Aaker (2004) acknowledges the importance of competition, noting that to complete an audit the analysis should consider competing brands in the product category. However, from a supply side perspective, competition is viewed as an attribute relative to each owned brand rather than as an element of market structure that may have intrinsic influence on the strategic choices resulting from an understanding of how consumers interact with the brand architecture.

Consumers do not always limit their choices to sets that are neatly defined in ways that make sense to brand managers and marketing executives. Customer perceptions, evaluations, and subsequent purchase behaviors often reflect a much messier market reality. The demand side perspective brings to bear on marketing strategy perceived relationships between brands and portfolios as well as competitive usage to satisfy variable needs. The process of assessing these relationships is rooted in the analysis of consumer data. For instance, a careful examination of data recording the selection and patterns of usage of competing brands may reveal opportunities for co-opetition (Nalebuff & Brandenburger, 1997) while most firm-based brand analyses only seek to identify the presence of complements and substitutions (Leitch & Richardson, 2003).

Approaching brand architecture from a consumer perspective draws a parallel to market structure analysis (MSA), a demand side approach to defining competitors and the bases of competition (Day, Shocker, & Srivastava, 1979; Srivastava, Alpert, & Shocker, 1984). The Aaker and Joachimsthaler (2000b) definition of brand architecture can be recast as a demand side concept that includes competitive brand relationships. Applying insights from MSA we can enhance the creation of brand architecture by incorporating the ways in which brands are used. Specifically, an analysis that includes both owned and competitive brands, and multiple usage categories, allows management to better understand its own brands. By isolating substitutes and complements from one brand to another, and across usage situations, managers can more precisely assign brand roles within the brand architecture.

Fully understanding brand architecture requires examining the market for competing brands, some of which are external to the company and some that might be part of the company's portfolio. The sizable literature on competitive MSA provides a foundation for this approach to brand architecture. MSA has examined the extent of competition between firms in an industry, resulting in the concept of strategic groups as a way to define the real set of competing firms (McGee, Thomas, & Pruett, 1995; Reger & Huff, 1993). MSA creates the foundation for interpreting competitive structure by identifying the similarities of product benefits and usage situations (Srivastava et al., 1984; Srivastava, Leone, & Shocker, 1981). Firms often rationalize brand architecture on the basis of the market segmentation of product and brand offerings in conjunction with the business objectives of the firm (Dabic, Schweiger, & Strebinger, 2004; Petromilli, Morrison, & Million, 2002), but ultimately the consumer by his or her purchasing and usage should be the arbiter of the company's brand architecture.

Brand architecture as a network of brands and usages

Brand architecture is predicated on deriving market structure by defining a set of relationships among brands. While the key strength of MSA lies in its conceptualization of markets as a set of products in different usage situations, branding gives added value to

products and drives purchases, with some brands offering more competitive value than others (Aaker & Joachimsthaler, 2000b). Similar to applying the MSA approach to products, it is useful to think of brands in conjunction with their usage in order to examine consumers' brand choice behavior. Toward this end, we demonstrate *network analysis* as an effective way to measure consumer behavior, capture brand interrelationships at a given level of the brand hierarchy, and derive a consumer-based view of each company's brand architecture.

Network analysis is predicated on graph theory, which identifies relationships between discrete objects. Networks have been applied to fields as diverse as physics, biology, organizational behavior, telecommunications, information science, and energy grid management (Barabási, 2003; Freeman, 2004). The linked objects, or network nodes, may be people, places, events, things, or ideas; they may even be combinations of these concepts, such as brands and usage situations. In social network analysis links between nodes are often derived from relationships such as friendship, communication, and exchange behavior. In typical marketing applications nodes may be organizational actors such as corporate officers, lobbyists, or doctors (Burt & Janicik, 1996), producers, suppliers, and consumers (Gulati, Nohria, & Zaheer, 2000; Martin & Clark, 1996), or brands (Henderson, Iacobucci, & Calder, 1998). When brands at a given level of brand hierarchy are used as nodes, the links between brands reflect some form of brand choice such as brand switching (Henderson et al., 1998; Marcati, 1996) or co-occurrence of purchase.

Generally, network theoretical constructs are associated with patterns or relationships such as the number of connections to an object or whether a pattern is repeated across the network. In order to capture the complete set of relationships necessary for describing a brand architecture we propose treating the set of owned (or controlled) and competing brands, and their uses, as a network consisting of all brand-usage situations (e.g. Ford for commuting vs. Ford for shopping vs. Toyota for shopping). The architecture for any given company and its brands is embedded in the network structure *and may include competitor brands*.

Network analysis of brands

Researchers have previously noted the utility of network analysis for understanding brand structures (Iacobucci, Henderson, Marcati, & Chang, 1996; Marcati, 1996). Henderson et al. (1998) have developed schema for applying various network properties such as centrality, cohesion, and position to the assessment of branding effects. By capturing the set of all possible brand interactions within a brand set, this analytic approach offers a parsimonious way of describing the competition between brands and provides an extensive set of derived measures that can be used to explore brand relationships (Wasserman & Faust, 1994). Henderson et al. (1998) propose that relationship variables and position characterizations created in a network analysis can offer precise operationalizations of brand features such as a driver brand (based on centrality), complement (based on cohesion), and cannibalization (based on relative position). All brands are potentially connected to all other brands through the processes of consumer evaluation, consideration, choice, purchase, and primary use. The approach taken in this article extends Henderson et al. and is consistent with MSA. That is, measure the pattern of brand purchases but include information about usage in order to describe the brands and their roles as a structure viewed from the consumer perspective.

Networks of brand-usage

The term *brand-usage* is used to describe nodes that represent brands conjoined to their usage situations. A network of brand-usages crosses all brands with all usages so that the

set of brands by usages represents all possible combinations that might occur in a market. It may be useful to think of brand-usage situations as sub-brands. The brand name is the same, but the context re-defines the brand in ways that are specific to the usage situation and the context in which it is used. By identifying co-occurrences of brand-usage nodes in individual consumers' purchase sets and applying network analytic procedures, our understanding of the complexity of brand architecture can be expanded. Network analysis provides a broad palette of techniques and constructs (Wasserman & Faust, 1994). In this application to brand architecture we have found three fundamentally useful ideas for developing marketing strategy: (1) types of linkages are critical to identifying structural relationships; (2) centrality identifies core brands and usage situations within the industry; and (3) nodal clusters (cliques or neighborhoods) identify key linkages for each individual brand-usage situation, which taken together give structure to a company's brand architecture.

Linkages and structural relationships

In a brand-usage network, each node contains two pieces of information, the brand used and the purpose for which it was used. Each dyad represents one of three possible situations: different brands with the same usage; the same brand with different usages; and different brands with different usages. When a consumer has multiple occurrences of the same brand with the same usage the result is a monadic rather than dyadic representation; monads identify repeat usage and behavioral loyalty to the brand when specific needs are addressed.

Dyads representing pairs of behaviors that occur with above average frequency indicate structural reasons for consistent behavior. For example, when the dyad consists of two nodes with the same brand but different usage behaviors, the pattern indicates consumer willingness to use one brand to address different needs or solve different problems. When the dyad consists of two nodes with different brands but the same application of those brands, the pattern indicates the two brands are considered acceptable substitutes for the usage situation. When neither the brand nor the usage is the same, but the nodes are nonetheless connected in the network structure, we have evidence of two brands being used to complement each other in order to address the multiple needs of consumers (Shocker, Bayus, & Kim, 2004). When dyadic co-occurrence is infrequent, the two brand-usage situations are not linked in the network and should be unrelated in the brand architecture.

Finding network patterns can help the marketer focus on building loyalty and avoiding specific competitive substitution. Attention to brand-usage dyads will identify when complementary usage stays within the company's portfolio or, alternatively, involves the use of a competitor's product. A company's marketing strategy may create a comprehensive and differentiated portfolio that covers all major segments of the market, or it may recognize that co-opting a competing brand for certain purposes is more advantageous than trying to build an alliance within the portfolio. For example, it may make sense to send a customer to a specific competitor when the cost of providing a product or service outweighs the benefits. This approach is frequently used by airlines when the cost of operating a route is not justified by the profit potential of the route.

Regardless of the intent or success of industry competitors to position their brands according to some strategically advantageous architecture, it is the market which determines how those brands are actually used. Thus the network of brand-usage patterns provides a foundation for developing, refining, and correcting a firm's brand strategy.

Centrality

Centrality is among the most fundamental network analysis statistics and signifies the position of one node relative to all others in the network. A brand-usage situation will tend to be central to the network when its node is frequently and therefore closely connected to other brand-usage situations. Nodes that are central to the overall network tend to be dominant brands. Often, but not always, the nodes most central to the network will have the greatest market share, be indicative of the company's leader brand(s), or identify what consumers perceive as the brand's most customary usage. If they are not dominant, they are likely to be a common link with two or more dominant brands. Using the network approach, it is possible for competitor brands to be viewed as part of the brand architecture, whether or not this is desirable. The appearance of a competitive brand-usage node may indicate a substitution or complement. If the competitor node is peripheral (not central), it is likely that consumers use the competitor brand to augment use of the owned/controlled brand. If the competitor node is central, consumers may be using the owned brand as an alternative to a dominant competitor. Brand centrality can change in response to market forces. As a consequence, the roles of competing brands may also change.

Nodal clusters

Observing clusters is also critical for the application of network analysis to brand architecture. Brand and usage patterns do not have to be limited to dyads. An individual's personal network of brand-usages may be dyadic, triadic, or of any rank that is one less than the number of brand-usage situations. At the individual level, we would not expect to find overtly complex structures since the number of dyadic brand-usage situations grows large quickly (with $(n \times n - 1)/2$ possible combinations).[1] The more brand-usage situations an individual experiences the more likely combining dyads into triads or more complex patterns will be idiosyncratic. But at the market level, frequently recurring patterns of brand-usage across consumers may form groups of dyads and provide clues for identifying important but more complex strategic brand interactions. For example, when both nodes of a dyad are part of multiple dyads, then a chain of co-occurrences may be formed. Thus the aggregation of dyads, chains, and other linkage arrangements found in personal brand networks creates a map or portrait of the overall brand architecture from a customer segment or total market perspective.

Cliques are groups of nodes that are completely inter-connected and form distinct clusters within the larger network structure (Luce & Perry, 1949; Wasserman & Faust, 1994). In social network analysis (from which the concept originates), cliques describe cohesive subgroups for which the linkages are relatively strong, direct, intense, frequent, or positive. In brand-usage networks, cliques form when brands and usages have frequent and persistent ties in the overall consumption framework. Most often, brand-usage cliques form around a single brand or a single usage. When a brand has high loyalty and customers choose that brand regardless of how they plan to use it, a brand clique is formed. When a functional need is satisfied by many brands interchangeably, a usage clique is formed.

Network neighborhoods and cliques are similar, though the former is typically used to describe patterns of regular equivalence. When two nodes have identical connections to a set of other nodes, even if all of the nodes in the set are not completely connected, the cluster may be more accurately described as a network neighborhood (Everett, Boyd, & Borgatti, 1990). Neighborhoods (sometimes called n-cliques) are a set of nodes with a high (but unlike cliques, less than 100%) interconnectedness (Wasserman & Faust, 1994).

Typically, the network analysis will focus on direct linkages between nodes, but it may be useful in some instances to consider indirect links; n-cliques allow for cliques to be identified when n indirect links are accepted for defining the clique. For example, if node A is connected to node B based on a common usage theme, the other connections to brand B (which are indirect connects to A through B) may also be based on the same usage. In this case, a likely interpretation is that B is the central node of a usage-based neighborhood and brand A is also a substitute for brands C, D, and so on.

Summary of the network approach

Purchase and usage data can capture the co-occurrence of brands (and their usage situations). Network analysis is a widely used tool for identifying and interpreting structural relationships that may be applied to these data in order to find brand structure in the market. Interpreting (1) the overall pattern of linkages, (2) central versus peripheral positioning of brand-usage nodes in the network, and (3) the clustering of brand-usage nodes yields information about how brands are related to each other within a portfolio or at one level of a brand hierarchy and, importantly, how their interactions with competing brands may influence the planning and adjustment of brand architecture. The following example provides insight into how this concept works in practice.

Car rental industry study

In order to demonstrate network analysis of brand-usage and its strategic application to brand architecture development, an analysis was conducted using data collected from car rental industry customers in the United States. The car rental industry provides a straightforward illustration because the number of major brands is limited and the number of car rental brands within each portfolio is small. While marketing strategists may be concerned with rationalizing their brand architecture across many product categories (as is the case with conglomerates such as Proctor & Gamble or Siemens), in practice diversification across unrelated industries will often provide the logic for major structural decisions within the corporate framework. The difficult decisions involve brands that sell overlapping goods and services. The car rental market provides a practical example of the kind of sub-structure analysis required for larger companies with many brands, but it does so by focusing on a situation comparable to those where only a few brands are likely to interact within the brand architecture. Equally important, the car rental study provides a clear illustration of how brands outside the strategist's control, the competitor brands, should be treated as structural elements and factored into the planning of a brand architectural for company-owned brands.

Consumer data

To demonstrate the strategic application of network analysis for operationalizing the brand-usage approach, this analysis focuses primarily on one company, Dollar Thrifty Automotive Group, Inc. (DTAG) and its two car rental brands, Dollar Rent A Car and Thrifty Car Rental. The choice of DTAG provides a useful illustration because its two brands are similarly positioned and the DTAG corporate platform relies on a functional structure rather than maintaining two separate brand marketing organizations within the company. Both brands share field operations, technology, and management. While transparent to customers who see Dollar and Thrifty operating as separate entities, the functional organization creates cost savings and operational efficiencies (DTG History,

2008). The decision to maintain two very similar brands supported by a common infrastructure raises interesting questions about brand architecture and the efficacy of DTAG marketing strategy.

In January 2004 a survey on car rental usage was sent to 75,000 US household members of the Synovate consumer mail panel. Synovate uses the balanced panel approach to match samples to the US Census demographics at the time of the survey. Matching is based on the characteristics of age and income of head of household, household size and composition, census region, and population density. A total of 40,602 completed surveys were returned (54.1%); 7171 households (18.7% of the completed surveys) reported renting a car in the prior 12 months and were included in the initial analysis. The 12-month period is used to account for seasonality in the travel industry; package goods may utilize shorter periods while durables may require longer duration for data aggregation.

The questionnaire was filled out by the household member with the most frequent car rental experience. In addition to determining whether or not the respondent household had rented a car in the prior 12 months, the core question of the survey asked respondents to fill in a grid indicating the number of rentals for each combination of brand and usage situation. The extent of rental activity was measured as the number of rentals (transactions), which provides a simple representation of the brand use; other measures of activity, such as the number of days rented or aggregate amount spent on car rentals in each brand-usage situation could also be used. The brand axis of the grid consisted of the eight leading car rental brands and an 'other' brand category. The usage axis of the grid consisted of five major usages; in the car rental industry usage is defined as type and location situations and includes rented for business purposes at/near an airport (BA), rented for business purposes at other locations (BO), rented for leisure/personal purposes at/near an airport (LA), rented for leisure/personal purposes at other locations (LO), and rented for insurance or replacement purposes (IR). The last category was not location specific; subsequent research has determined that insurance and replacement rentals (which are associated with the maintenance and repair of owned vehicles) rarely occur at airport rental locations. The resulting grid yields 45 brand and usage combinations; each respondent's grid provides an individual profile of annual car rental usage frequency for the 45 categories. Most renters reported using only a small subset of the 45 brand-usage categories.

Market share

Results of the January 2004 survey indicated that in the US car rental industry Enterprise (25.8% of transactions) and Hertz (18.9%) brands were the market leaders (Table 1). At the time of the study, the eight major brands were owned by five companies. Enterprise was a privately owned stand-alone brand. Hertz was a single car rental brand within the portfolio of brands owned by Ford Motor Company. Avis (14.4%) and Budget (10.5%) were both components of a travel and real estate portfolio owned by Cendant Corporation. National (8.7%) and Alamo (5.1%) comprised a two brand portfolio which had emerged from bankruptcy in 2003 as Vanguard Car Rental USA Inc. (owned by capital investment firm Cerberus and subsequently sold to Enterprise in 2007). Dollar (4.0%) and Thrifty (3.4%) formed the two brand portfolio of Dollar Thrifty Automotive Group Inc. Collectively, minor car rental brands and independent/local operators accounted for 9.2% of rental transactions.

Positioning strategies used by these firms were inferred from a supply side audit of the contents of annual reports, website information, and industry analysts' reports. These positions were compared to the actual usage data reported from the consumer survey to

Table 1. US car rental market share (% of total transactions in 2003).

Company and brand	Company rentals (%)	Brand rentals (%)
Enterprise	25.8	25.8
Cendant	24.9	
Avis		14.4
Budget		10.5
Hertz	18.9	18.9
Vanguard	13.8	
National		8.7
Alamo		5.1
DTAG	7.4	
Dollar		4.0
Thrifty		3.4
All other	9.2	9.2
Total	100.0	100.0

qualitatively assess the fit between strategies and outcomes (Table 2). In general, supply side positioning strategies would appear to be confirmed, but they do not tell us much about the interaction of brands. An approach that applies brand and usage information simultaneously, and does so across customers, has the potential to yield much greater insight.

Network analysis of US car rental data

Network analysis procedures involve transforming the data to a representative form for deriving and examining structure. The structure can be interpreted visually and with the use of diagnostic measures that quantify characteristics of the structure. This analysis of US car rental data has three components: identification and categorization of the structural linkages in the brand-usage network; calculation of centrality or importance scores for each of the brand-usage nodes; and identification of cliques and neighborhoods, which can be used to specify the set of nodes and linkages for any individual brand's architecture.

Table 2. Comparison of brand strategy to usage mix (% of transactions within brand in 2003).

		Mix of brand-usage types (%)					
Brand	Positioning	BA	BO	LA	LO	IR	Total
Enterprise	BO/LO/IR	12.2	15.6*	14.6	28.5*	29.1*	100.0
Avis	BA/LA	54.7*	6.4	27.6	8.7	2.5	100.0
Budget	LA/LO	38.2	5.0	42.2*	11.6	3.0	100.0
Hertz	BA/LA	58.9*	3.8	29.7	4.8	2.8	100.0
National	BA	56.5*	4.3	33.4*	3.2	2.5	100.0
Alamo	LA	34.5	1.3	56.0*	5.0	3.2	100.0
Dollar	LA	37.9	2.4	51.2*	4.7	3.7	100.0
Thrifty	LA	33.0	5.8	47.5*	9.2	4.5	100.0
All other	BO/LO/IR	19.2	15.5*	29.0	22.9*	13.4*	100.0
Industry average		37.6	8.3	30.0	13.7	10.4	100.0

Note: Brand-usage types: BA = business rental at/near airport; BO = business rental at other locations; LA = leisure/personal rental at/near airport; LO = leisure/personal rental at other locations; and IR = insurance/replacement rental.
*$p < .05$ and above the industry average for this type of brand-usage.

Data transformation

A high proportion of those renting a car in the past year (59.9%) rented two or more times for any use. Of the 7171 respondents, 2878 had only one rental; by definition, these respondents had no co-occurrences and for this analysis were eliminated from the data set. Among the remaining respondents, 1025 had multiple rentals involving single-brand/single-usage (unconnected) situations; these were also excluded.

Using the grid question, respondents not only indicated the brands used but their purpose for renting the vehicle and whether that rental occurred at or near an airport or at some other location not associated with air travel. This is a critical distinction because, as Srivastava et al. (1984, p. 32, emphasis added) note, 'over a period of time customers may develop sets of *products* for consideration based upon the perceived appropriateness of their functional attributes for the intended usage.' Generically, car rental is a product that has typical functional uses and occurs at typical locations; any of the brands could serve any of these functions and be provided at any of these locations. However, the preferred uses and locations that become associated with each brand define those brands' appropriate functional attributes from a consumer perspective.

The combination of nine brands and five usage situations yields 45 nodes. Each node can be considered a separate product alternative, the application of a brand to meeting a specific consumer need or the solution to a specific consumer 'problem'. The 45 nodes have $(n \times n - 1)/2$ or 990 potential linkages; the links represent the co-occurrence of brand-usage situations in each consumer's purchase set.

The data yield a two-mode matrix of respondents by brand-usage nodes. The final data matrix includes 45 brand-usage situations by 3268 respondents. Figure 1 illustrates four categories of brand-usage patterns.

Pattern A is single brand/single usage and includes both one-time renters (during a 12-month period) and those with attitudinal or behavioral loyalty over multiple rentals. Pattern B, multiple brands/single usage, is the classic brand substitution condition while Pattern C, single brand/multiple usage, is indicative of brand loyalty across differing needs. We expect to see Pattern B when consumers do not perceive major vehicle, service, or reputational differences across brands. If brands are distinct, then we expect Pattern C. Pattern D, multiple brands/multiple usage, is the most complicated in that it can take various forms. This may occur when two (or more) brands are used as complements to

		Number of brands used	
		One	Two or more
Number of usage situations	One	A Singular pattern	B Substitution pattern
	Two or more	C Loyalty pattern	D Complexity pattern

Figure 1. Four primary types of brand-usage situations.

address different needs, but it may also mask a complex set of loyalty and substitution issues that are impacted by market forces such as availability at a particular location, price competition, or a company's portfolio strategies (see Day et al., 1979 and Shocker et al., 2004 for discussion).

In the two-mode affiliation matrix of brand-usage situations by respondents, each cell represents an individual's frequency of a brand-usage situation occurring in the past 12 months. In order to interpret market structure, the two-mode affiliation matrix was transformed to a one-mode, square symmetrical brand-usage by brand-usage similarities matrix of Pearson correlation coefficients (PCC). Market structure is defined by structural equivalence, or the set of roles and positions represented by the brand-usage nodes. There are many alternatives for determining structural equivalence; the proximity measure may be dichotomous or valued, directional or non-directional, and including or excluding self-ties from the matrix diagonal. In a valued structure such as the car rental data, proximity is typically represented as Euclidean distance, covariance, or correlation (cf. Wasserman & Faust, 1994). Since the presence or absence of relationships is based on dichotomizing the valued data, we chose to use the standardized form of valued similarity data, the Pearson correlation coefficient.

The 990 pairs in the PCC matrix have a mean of 0.0162 and standard deviation of 0.051. The proximity, or similarities, matrix was dichotomized by assigning those similarities which exceed two standard deviations from the mean similarity score a value of 1 and all other similarity scores a value of 0 (Figure 2). The choice of 2 SD approximates the cut level for a scree test of the distribution of PCC values. While every dyad (or pair of nodes) has a score, the majority of these scores are small enough that their co-occurrences are largely idiosyncratic, and in the network structure these dyads are represented as unlinked. Pairs occurring with sufficient frequency that they cannot be readily attributed to chance are represented in the network structure as the linked nodes.

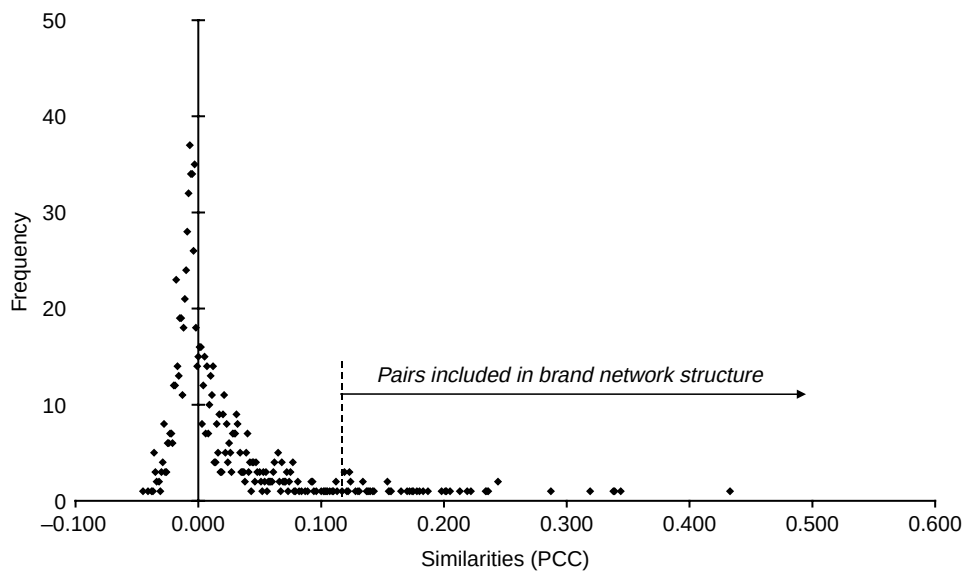

Figure 2. Frequency distribution of similarity scores from the 45×45 car rental brand-usage situation adjacency matrix ($n \times (n - 1) = 990$ dyads). Dyads with similarity scores equal to or greater than two standard deviations ($r = 0.102$) are treated as linked for the network analysis.

Network structure

The 45 nodes and 990 potential links in the car rental data, when constrained to similarity levels of two standard deviations or more above the mean, yield 41 nodes connected by 53 links. Overall this is a relatively sparse network (density = .0535 or 53 observed links/990 potential links). Four nodes are classified as isolates; these are brand-usage situations that tend not to occur in conjunction with other car renting behaviors and, coincidently, have relatively low market share. The 53 links represent three kinds of rental patterns; 31 links (58.5%) join nodes with a common brand (Pattern C), 15 links (28.3%) join nodes with a common usage situation (Pattern B), and seven links (13.2%) join complementary nodes (Pattern D, different brand and different usage situation). The remaining 937 potential links had very small PCC values and were dichotomized with a value of 0; although these small values can indicate some co-occurrence, the links are considered to be idiosyncratic. Of the 53 linkages, only one joins nodes that are part of the same firm's brand portfolio; among airport renters, Dollar and Thrifty are substitutes for each other when used for leisure/personal travel (shared usage link).

The market structure was mapped by applying algorithms and visualization with the UCINET 6 network analysis program (Borgatti, Everett, & Freeman, 2002). Figure 3 shows the nodes grouped into linked clusters. The node symbols indicate type of usage for the labeled brand, the connecting lines indicate the linkage type (Pattern B, C, or D), and the size of the node symbol is proportional to market share. The clusters are defined on the basis of weak ties (Granovetter, 1973); when a group within the network is connected to another group through one (or few) ties, the groups are considered distinct subsets. Four groups emerge in the US car rental market: (1) the largest, core cluster is dominated by Hertz and Avis, and includes Enterprise for non-airport use; 'other brands' nodes are also part of this group as a result of substitution for the major brands (typically when they are not available in secondary and tertiary airport markets). (2) A second, alternative brand cluster, includes most National and Budget nodes and two Dollar non-airport nodes. (3) The third cluster is a discount grouping of low-cost brands used primarily for airport rentals. (4) The fourth cluster consists of Thrifty brand-usages and one Dollar node substituting for Thrifty leisure airport rentals. This network structure provides an overall portrait of the US car rental market, suggesting that the market is segmented by both brand dominance and usage factors, and provides a foundation for identifying individual brands' and companies' brand architectures.

What is immediately apparent for DTAG is that the Thrifty renters tend to use Thrifty for multiple purposes and, given market share, constitute a small but generally cohesive group whose repeated Thrifty usage across situations indicates loyalty. Conversely, Dollar has no direct linkages between any of its five brand-usage nodes. Instead, Dollar nodes are appended to competitor nodes and are in all cases brand substitutes in specific usage situations. Given the consistency of this pattern, Dollar is likely to be the secondary brand in most substitution situations.

Network centrality

Network centrality is an index of the tendency for some nodes to exist at the core of the structure while others reside along the periphery. Brand-usage nodes in the center may be more important, may play specific roles relative to the surrounding nodes, may identify leader brands along with their primary usage(s), or have other qualities that distinguish them from the majority of nodes in the network. In the brand-usage market structure, central nodes tend to account for more market share or are more frequently substituted or

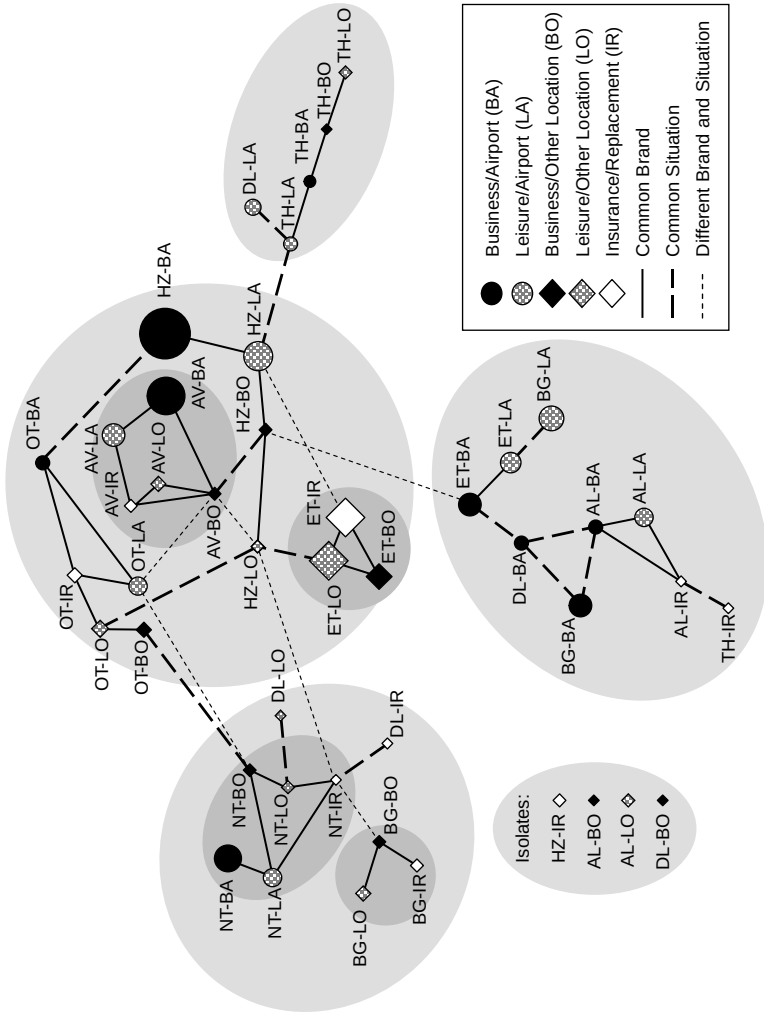

Figure 3. Brand-usage network structure for the US car rental industry.
Notes: The first two letters of each label indicate brand. The size of the node symbol is proportional to market share for the brand-usage situation. Background shading indicates major and minor sub-divisions of the network structure; the four major divisions indicate usage clusters that are relatively distinct from each other.

have desirable product attributes that encourage one brand's use in conjunction with other brands. Nodes on the periphery will be isolates or connected by a single linkage (pendants) or may be the end of a chain of single linkages. In the market structure, peripheral brand-usage situations resulting in substantial market share will be infrequent, atypical, or both. When this does occur it may reflect a strategic choice; for example, a niche brand may be very successful but have relatively little connection to other brands and usage situations (cf. Iacobucci et al., 1996 and Marcati, 1996, in which the successful BMW automotive brand is positioned as an isolate in networks based on brand-switching data). Centrality in a brand-usage network can be defined in several ways (Kilduff & Tsai, 2003; Serota & Bhargava, 2006; cf. Wasserman & Faust, 1994, for computational formulae). These include the number of links one node has to other nodes (degree centrality), the reciprocal of the number of links necessary to connect one node to all other nodes (closeness centrality), the extent to which one node is interposed on all paths between all other nodes (betweenness centrality), and a normalized factor score indicating the extent to which central nodes are connected to other central nodes (eigenvalue centrality). Whether or not a brand is central or on the periphery is a key to understanding the extent to which competition, or even other brands within the portfolio, must be accounted for strategically.

As Table 3 shows, the four measures are not perfectly correlated but collectively give a good indication of the most central brand-usage nodes. The most central brand-usage nodes are Hertz non-airport business and leisure rentals and Avis non-airport business rentals. All three nodes are highly connected, are close to and located along many paths between other nodes, and tend to be linked to other central nodes. None of the three most central nodes accounts for a large market share but their importance is a result of being included in the rental activity of diverse groups of consumers. They are conduits by which brands interact. Seven of the 10 most central nodes are part of the core market segment.

Table 3 also provides centrality measures for each of the DTAG nodes and each of the competitor nodes with a link to DTAG. Both DTAG brands are outside of the central core of the network. Although not statistically significant, the mean of the 10 DTAG nodes for each of the centrality measures is directionally lower than the mean for the 45 network nodes. However, two of these nodes indicate centrality in the network sub-structures. In the discount cluster, Dollar used for airport business rentals has degree = 3 and above average betweenness = 175.0, making it the central brand-usage of the cluster (Enterprise has higher connectedness due to its position linking the discount cluster to the core cluster). All of the Dollar business airport links are to other brands with the same usage, Alamo, Budget, and Enterprise. The Dollar node has the smallest market share among the four; the combination of position, centrality, and market share point to Dollar as the 'go-to' substitute rental brand among business renters seeking a low-cost airport rental. Thrifty used for leisure airport rentals also has high centrality relative to the other nodes in the stand-alone Thrifty brand cluster (degree = 3 and betweenness = 147.0). This node appears to represent the dominant usage for the Thrifty brand and links Thrifty to Dollar and Hertz as a substitute for leisure travel use.

Nodal clusters

Network neighborhoods are patterns or clusters of nodes within the larger network structure. Cliques are a form of neighborhood and include groups of nodes in which all members of the clique are connected to all others within the clique. Cliques provide a fine-grained picture of brand strength by highlighting the substitute and complementary brand-usage within the larger network structure. The car rental data creates a fairly sparse

Table 3. Centrality measures for leading brand-usage nodes, DTAG nodes, and nodes connected to DTAG.

			Centrality			
Rank	Brand	Usage	Degree	Closeness	Betweenness	Eigenvalue
Most connected brand-usage nodes						
1	Hertz	LO	5	15.02	315.00	0.39
2	Avis	BO	6	14.67	190.00	0.44
3	Hertz	BO	4	15.12	398.50	0.32
4	National *	IR	5	14.01	242.83	0.23
5	Hertz *	LA	4	14.19	222.83	0.17
6	Other	LA	4	13.75	77.83	0.28
7	Enterprise *	BA	3	14.01	268.00	0.12
8	National	BO	4	13.02	47.67	0.19
9	Other	LO	3	13.67	42.67	0.20
10	Enterprise	LO	3	13.71	41.00	0.17
DTAG nodes and competitor nodes connected to DTAG						
12	Dollar	BA	3	12.83	175.00	0.05
13	Thrifty	LA	3	12.90	147.00	0.06
17	National *	LO	3	12.90	47.00	0.13
20	Alamo *	BA	4	11.73	111.00	0.03
28	Thrifty	BA	2	11.70	76.00	0.02
29	Dollar	IR	1	12.47	0.00	0.07
30	Alamo *	IR	3	10.71	39.00	0.01
31	Budget *	BA	2	11.64	0.00	0.02
32	Thrifty	BO	2	10.65	39.00	0.01
34	Dollar	LO	1	11.58	0.00	0.04
35	Dollar	LA	1	11.58	0.00	0.02
40	Thrifty	IR	1	9.78	0.00	0.00
41	Thrifty	LO	1	9.74	0.00	0.00
45	Dollar	BO	0	na	0.00	0.00
Mean			2.36	12.54	61.40	0.10
SD			1.39	1.28	94.81	0.11

Notes: * Competitor brand-usage nodes connected to one or more DTAG nodes. Brand-usage nodes are ranked on the average Z-score for four centrality measures.

network; its 53 links are only 5.4% of the possible linkages. As a result we observe only six complete cliques, each of which is a triad. In four of these, the triad consists of three usages of the same brand (Avis, Enterprise, Alamo, and 'Other'). One usage clique emerges, airport business rentals from Dollar, Budget, and Alamo. The remaining clique consists of the three most central brand-usages in the network, Hertz for both business and leisure rentals at non-airport locations and Avis for business rentals at non-airport locations (the implication is that Hertz and Avis compete for primary airport usage but are seen as substitutable for secondary, non-airport rentals).

Another form of neighborhood is the ego-net, the set of nodes with links to one specific node (ego); dyads in which one node has a single linkage to another node are the smallest form of ego-net. In the market structure, these might occur when a brand has a frequent secondary use by customers who use it most often for another purpose. If a brand is used in a variety of ways by the same customers, the ego-net may have multiple linkages in which the brand-usage involves the same brand but different usage situations; and if another brand tends to be used in conjunction with the first brand (i.e. as a substitute or complement), the ego-net may expand across brands. Including all of the brand-usage

linkages for a single brand is another way to portray the network neighborhood. This can result in meaningful patterns of competitive interaction. It may also result in multiple neighborhoods when the network is analyzed at the brand, or even at the firm level. Figure 4 shows DTAG, which includes all uses of the Dollar and Thrifty brands, consisting of three separate components or neighborhoods within the car rental market structure (and one isolate). One component consists mainly of multiple usages of the Thrifty brand; a second component consists primarily of Dollar linked to other low-price brands competing for airport rentals, and the third group shows Dollar linked (and competing with) National at non-airport locations.

Whether or not the identified DTAG linkages are the desired linkages is an issue for DTAG management; the structure informs brand management that the two-brand portfolio has three key positions based on customer usage. Customer usage indicates that the Dollar brand serves primarily as an alternate or substitution brand; the specific competitors depend on the usage situation. These same data show that Thrifty customers use the brand for multiple purposes; the pattern suggests a small but brand loyal customer base.

Demand-based brand architecture

The network examination of structural relationships and key diagnostics shows far more complexity than might be found in the traditional organizational chart of brands. If the network is reduced to the individual brands, the US car rental industry appears dominated by companies employing a house of brands portfolio strategy (the exception at the time these data were collected was Enterprise Rent-A-Car, which was a stand alone brand that applied the company brand to all component products; Enterprise has subsequently become a house of brands that includes National, Alamo, and the WeCar car sharing service). Applying the brand-usage structure, the brand names may be the same, but the brands fulfill different roles and interact with different competitive sets in ways that may not be apparent from the brand-level definition of structure. Viewed as a house of brands, DTAG appears to be a very simple structure with two poorly articulated brands that share a single strategy. As part of a larger network that is defined on the basis of usage as well as branding, the multi-brand strategy has logical differences (we make no judgment about the correctness of this approach; but the network analysis demonstrates that it exists).

Visualizing the DTAG nodes and related network nodes as a hierarchy of sub-brands shows clearly that the two brands are used differently, and in ways that are defined by the competitor brand in each usage situation (Figure 5). Dollar, when considered in conjunction with each of its usage situations, is positioned to provide a low-cost substitute for other, higher volume brands. In general, customers who rent from Dollar for one purpose do not rent from this brand for other purposes. Dollar has a small market share and is positioned to be an available alternative when the preferred brand is not available in the desired location, is out of cars to rent, or when Dollar's rates are more attractive. The apparent strategy is to be inexpensive and available.

Supply side auditing indicates that like Dollar, Thrifty also follows a low-cost strategy. However, brand-usage data indicate that the brand is far more cohesive; Thrifty car renters are likely to use Thrifty for multiple purposes. If not doing so already, DTAG brand management could rationalize this brand architecture by focusing on the Dollar brand purely from a price-advantage perspective while developing Thrifty as the strategic center of the firm, combining favorable pricing with loyalty-building actions to achieve best-cost/value positioning. This strategy is augmented by competitive positioning. Thrifty has

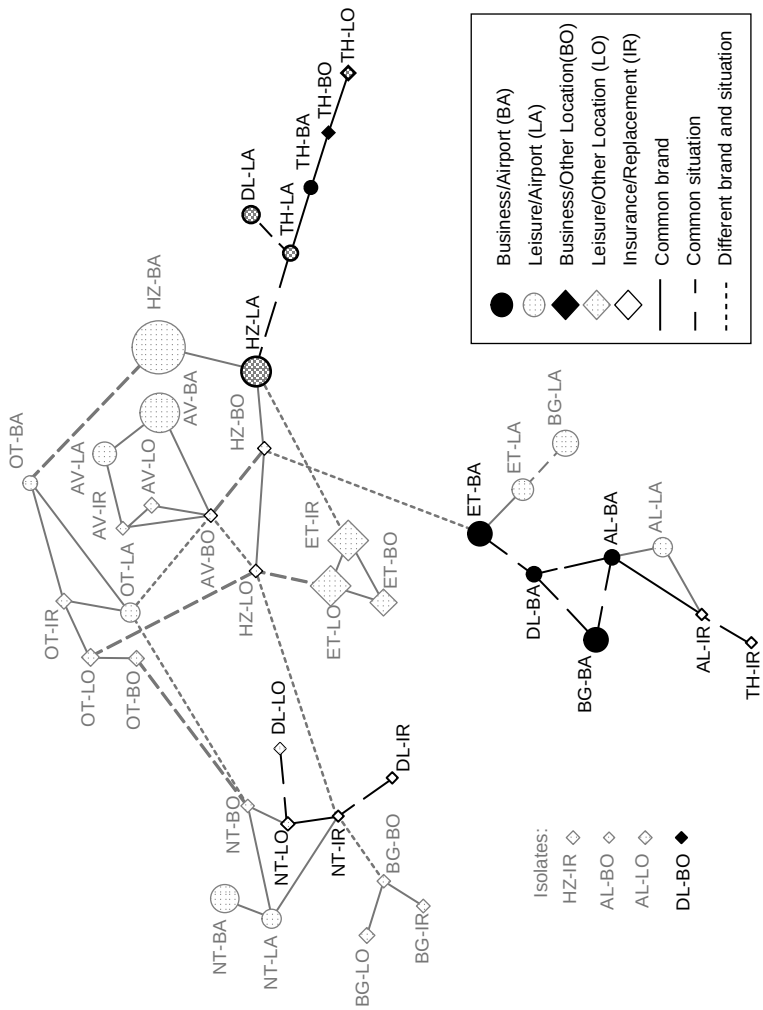

Figure 4. DTAG network neighborhoods within the brand-usage network structure. The neighborhood is defined as all direct links and n = 1 indirect links to all Dollar and Thrifty brand-usage nodes.
Notes: The first two letters of each label indicate brand. The size of the node symbol is proportional to market share for the brand-usage situation. Bold symbols and lines indicate nodes and links in the DTAG network neighborhoods. Shaded lines markers and lines are not connected to DTAG.

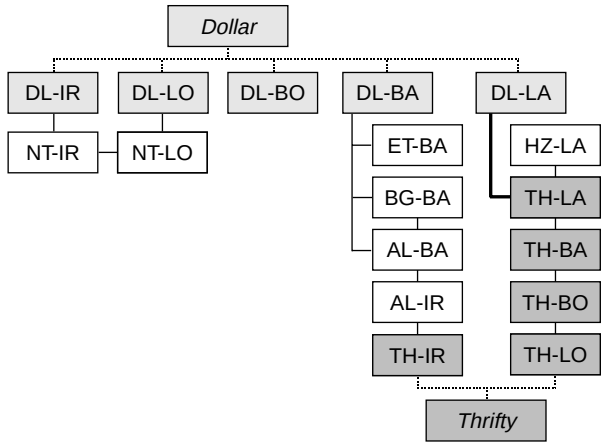

Figure 5. DTAG brand architecture derived from the network structure.
Notes: The first two letters of each label indicate brand. Solid lines indicate network linkages. The heavy solid line designates a network linkage between DTAG brands for leisure/personal usage situations.

only one substitution link in the network; in the area of leisure travel, Thrifty substitutes for Hertz, one of the two dominant brands in the market.

Discussion

More than two decades ago a series of articles (Day et al., 1979; Srivastava et al., 1981, 1984; Urban & Hauser, 1980) provided a foundation for a customer-oriented approach to understanding market structure. One of the main themes in this development was the importance of situation, or product usage, taken in conjunction with other customer and product variables. Since then, a large body of literature citing this approach has emerged. By far, the main focus of this literature has been the inclusion of situational factors for market segmentation. Research to help organize and delineate usage situations has employed a variety of popular methodologies, including hierarchical cluster analysis, multidimensional scaling, and conjoint modeling.

A separate literature has emerged to describe the importance of brand architecture for the effective implementation of marketing strategy. To a large extent theories of branding are rooted in supply side experience and the observation of successful marketing practice, including the implementation of different archetypal portfolio styles. In an effort to better understand brand structures and find a method to quantify the influence of competing brands on individual firm's branding decisions, it seemed clear that the same consumer orientation used by Srivastava and his associates to describe product-markets could be used to capture brand structures. Similar to Srivastava et al.'s (1984) finding that the interaction of product and usage could enhance market segmentation, we note that brand and usage interaction can more completely describe the perception and influence of brand architectures from the consumer perspective.

Limitations

We have demonstrated how the basic application of network analysis can enhance the understanding of brand architecture. Many issues remain. Internal to the methodology is

the exploration of different data and analytic forms. Our approach focuses on the presence or absence of links, but network techniques can account for the value of link strength and the presence of asymmetrical links (e.g. a primary brand's links to subordinate or 'backup' brands). By observing the order in which purchase events occur, measuring brand preferences, and identifying primary usage needs it would be possible to more conclusively establish dominance in each of the brand-usage situations. Furthermore, we have only hinted at the dynamic character of networks. We can hypothesize about how markets change as brands erode, new brands are created, and mergers and acquisitions alter brand portfolios. While the present article is illustrative, these hypotheses need to be tested. Measuring causal influences, or inducing specific marketing actions, and observing change in the network structure is the *sine qua non* for those in brand management to accept the brand-usage conceptualization of brand architecture.

Future research should also look at traditional metrics in conjunction with brand-usage. Measures of brand awareness can be constrained by usage situations in order to better understand the formation of competitive sets. Measures of brand imagery, purchase reasons, and evaluative priorities can also be constrained to individual usage situations. Purchase intentions can provide a dependent measure for predicting future activity as a consequence of individual usage behaviors.

Methodologically, it may be possible to refine brand-usage networks by making different choices among the plethora of network techniques. As we noted in the description of our analysis for this study, we chose to use correlations as our measure of proximity and as the basis for dichotomizing the data and identifying structural linkages. While the approach is parsimonious, we need to ask whether or not information would have been preserved if we had chosen to use the covariance matrix rather than the correlation matrix. These kinds of issues have been explored in depth in the application of networks to social relationships but need to be revisited in the context of market structure and brand architecture.

Conclusion

To capture brand architecture using a demand side approach, we wanted a technique that would preserve and display the complexity of brand and usage interactions rather than summarize or categorize them according to the methods employed by most segmentation procedures. Iacobucci (1996) and her colleagues demonstrated the value of network analysis to analyze consumer behavior in their examination of brand switching data. Furthermore, they showed that the network approach was not only complexity-preserving; it also has the capability to be adaptive and dynamic. Beyond this initial exploration of marketing networks, the broader fields of physical and social network analysis provide an entire body of structural constructs which can be meaningfully extended to the analysis of brand, situation, and consumer interactions. While the previous applications of network analysis to market structure have broadened the marketing lexicon, the application of network analysis to brands in multiple usage situations suggests a rich new approach to quantifying brand interactions, enhancing marketing strategy, and managing brand architecture.

This article reconceptualizes brand architecture as the set of brand and usage relationships that can be derived by examining consumer purchase behavior. A clear extension of this work would be to examine the changes in the brand architecture over time to capture the effects of endogenous variables and identify exogenous influences. From this perspective, brand architecture provides insights about the resilience of brands as well as their flexibility when used in multiple situations. The categorization of brands in Figure 1

provides a useful starting point. Clearly structural relationships among brand-usage situations will migrate over the four cells, some because of repositioning and resource allocation, others because of competitive situations or changing consumer needs. Such analyses will not only be useful to marketing managers but also provide a bridge to addressing the capabilities that strong brands represent. Used this way, brand-usage networks provide a framework for making brand architecture quantitative, consumer-driven, dynamic, and actionable.

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Note

1. There are exceptions. In some industries, especially package goods, where consumers make frequent and multiple purchases, triadic and higher order interactions can be common. For example, when people shop for cereal, certain complex patterns of brand-usage patterns might be expected: Quaker Oats for mom and dad, Kellogg's Frosted Flakes for the kids, and General Mills Rice, Corn, and Wheat Chex to make Chex mix for a party. At least among the most popular brands, it might not be unusual to find these combinations or minor variations (structural equivalents) recurring. In industries where purchasing is less frequent, such as appliances and other consumer durables, triadic and higher order interactions might be rare or, difficult to observe.

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